

Exp no:4

Querying Ledger Data with Hyperledger Fabric

Aim:

To query and retrieve ledger data from a Hyperledger Fabric blockchain network using JavaScript chaincode, and verify asset information stored in the world state database.

procedure:

1. Navigate to the test network

None

```
cd ~/fabric-dev/fabric-samples/test-network
```

2. Start the Fabric network

None

```
./network.sh up createChannel -ca
```

```

/home/ram/fabric-dev/fabric-samples/test-network/./bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/mychannel.block -channelID mychannel
2026-09-08 17:36:36.935 IST 0001 INFO [common.tools.configtxgen] main → Loading configuration
2026-09-08 17:36:36.938 IST 0002 INFO [common.tools.configtxgen.localconfig] completeInitialization → orderer type: etcdraft
2026-09-08 17:36:36.938 IST 0003 INFO [common.tools.configtxgen.localconfig] completeInitialization → Orderer.EtcdRaft.Options unset, setting to tick_int
erval:"500ms" election_tick:10 heartbeat_tick:1 max_inflight_blocks:5 snapshot_interval_size:16777216
2026-09-08 17:36:36.938 IST 0004 INFO [common.tools.configtxgen.localconfig] Load → Loaded configuration: /home/ram/fabric-dev/fabric-samples/test-networ
k/configtx/configtx.yaml
2026-09-08 17:36:36.939 IST 0005 INFO [common.tools.configtxgen] doOutputBlock → Generating genesis block
2026-09-08 17:36:36.939 IST 0006 INFO [common.tools.configtxgen] doOutputBlock → Creating application channel genesis block
2026-09-08 17:36:36.939 IST 0007 INFO [common.tools.configtxgen] doOutputBlock → Writing genesis block
+ res=0
Creating channel mychannel
Adding orderers
+ . ./scripts/orderer.sh mychannel
+ '[' 0 -eq 1 ']'
+ res=0
2026-09-08 17:36:25.353 IST 0001 INFO [channelCmd] InitCmdFactory → Endorser and orderer connections initialized
Error: proposal failed (err: bad proposal response 500: cannot create ledger from genesis block: ledger [mychannel] already exists with state [ACTIVE])
Status: 405
{
  "error": "cannot join: channel already exists"
}
Channel 'mychannel' created
Joining org1 peer to the channel ...
Using organization 1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=1
2026-09-08 17:36:52.195 IST 0001 INFO [channelCmd] InitCmdFactory → Endorser and orderer connections initialized
Error: proposal failed (err: bad proposal response 500: cannot create ledger from genesis block: ledger [mychannel] already exists with state [ACTIVE])
After 5 attempts, peer0.org1 has failed to join channel 'mychannel'

```

3. Deploy JavaScript chaincode

None

```
./network.sh deployCC \  
-ccl javascript \  
-ccn basic \  
-ccp ../asset-transfer-basic/chaincode-javascript
```

4. Set Fabric environment variables

None

```
export PATH=${PWD}/../bin:$PATH  
export FABRIC_CFG_PATH=$PWD/../config/  
  
export CORE_PEER_TLS_ENABLED=true  
export CORE_PEER_LOCALMSPID="Org1MSP"  
  
export  
CORE_PEER_TLS_ROOTCERT_FILE=${PWD}/organizations/peerOrganizations/  
org1.example.com/peers/peer0.org1.example.com/tls/ca.crt  
  
export  
CORE_PEER_MSPCONFIGPATH=${PWD}/organizations/peerOrganizations/  
org1.example.com/users/Admin@org1.example.com/msp  
  
export CORE_PEER_ADDRESS=localhost:7051  
  
export  
ORDERER_CA=${PWD}/organizations/ordererOrganizations/example.c  
om/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.c  
om-cert.pem
```

5. Initialize the ledger

None

```
peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile "$ORDERER_CA" \  
-C mychannel \  
-n basic \  
-c '{"function":"InitLedger","Args":[]}' \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles  
"${PWD}/organizations/peerOrganizations/org2.example.com/tlsca  
/tlsca.org2.example.com-cert.pem" \  
--waitForEvent
```

6. Query all assets

None

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["GetAllAssets"]}'
```

You should see the six initialized assets:

None

```
asset1  
asset2  
asset3  
asset4  
asset5
```

```
asset6
```

7. Query a specific asset

None

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset1"]}'
```

8. Create a new asset

None

```
peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile "$ORDERER_CA" \  
-C mychannel \  
-n basic \  
-c \  
'{"function":"CreateAsset","Args":["asset20","Green","8","Arun",  
"1500"]}' \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles \  
"${PWD}/organizations/peerOrganizations/org2.example.com/tlsca/  
tlsca.org2.example.com-cert.pem" \  
--waitForEvent
```

The manual specifies `asset20`, Green, size 8, owner Arun and value 1500.

9. Query asset20

None

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset20"]}'
```

Expected:

None

```
{  
  "ID": "asset20",  
  "Color": "Green",  
  "Size": 8,  
  "Owner": "Arun",  
  "AppraisedValue": 1500  
}
```

10. Query all assets again

None

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["GetAllAssets"]}'
```

Now the result should include:

None

```
asset1
```

```
asset2
asset3
asset4
asset5
asset6
asset20
```

11. Query a non-existing asset

This demonstrates how Fabric handles a missing record.

None

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset50"]}'
```

```
(ram@ram)-[~/fabric-dev/fabric-samples/test-network]  
$ peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset50"]}'  
Error: endorsement failure during query. response: status:500 message:"The asset asset50 does not exist"
```